

Qusay Qadir

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EXPERIENCE

Borealis AI - RBC

Jan 2026 – Present

Software Engineer Intern

Toronto, ON

- Developed an **NLP risk platform** for Lumina Agentic Gateway, classifying prompts and tool permissions across **100+ production AI agents** and cutting deployment approval from **2 weeks to 3 days**.
- Engineered a **FastAPI** alerting service supporting **2K+ monthly notifications** across **12+ platform teams** with PostgreSQL, retries, audit trails, and Temporal workflows.
- Engineered Lumina Portal's onboarding workflow engine in **Go**, persisting Temporal state to PostgreSQL and exposing authenticated asynchronous REST APIs for **200+ monthly submissions**.

Homewise.AI

July 2025 – Dec 2025

Software Engineer Intern

San Francisco Bay Area, CA

- Architected an agentic request-routing layer in **Python** with Google ADK and MCP, dispatching queries to listing search, document RAG, and deterministic database tools across **2K+ listings**.
- Optimized session and query caching with **GCP PostgreSQL**, increasing cache hit rate from **20% to 88%** and reducing repeat-query latency from **~3 min to seconds**.
- Built a golden-dataset evaluation and prompt-versioning pipeline gating releases, integrating **Vertex AI Vector Search** and improving recommendation relevance by **55%**.

McMaster Exoskeleton

Nov 2024 – Dec 2025

Software Engineer

Hamilton, ON

- Accelerated multi-GB dataset uploads by **40%** through multithreaded, checksum-verified ingestion using **AWS S3, Lambda, and Spark**.
- Developed and evaluated a **TensorFlow** model for real-time movement estimation from exoskeleton sensor data, achieving **6.2%** mean prediction error.

Scotiabank

Apr 2024 – Aug 2024

Software Engineer Intern

Toronto, ON

- Migrated **4TB+** of IBM mainframe data through a **PySpark, Dataproc, and Airflow** ELT pipeline into BigQuery, reducing analyst query latency by **30%**.
- Developed **unit and integration tests** and GitHub Actions CI/CD pipelines gating dev/prod releases, managing environment-specific secrets and automated release validation.

TECHNICAL SKILLS

Languages: Python, Go, SQL, Java TypeScript, C++

Backend/Systems: FastAPI, Temporal, REST APIs, Redis, LangGraph, MCP

Data/Databases: PostgreSQL, MongoDB, BigQuery, Spark, Airflow

Cloud/DevOps: AWS, GCP, Docker, OpenShift (Kubernetes), Git/GitHub, GitHub Actions

PROJECTS

F1 Telemetry Intelligence Engine | Python, TypeScript, AWS, LangGraph

Mar 2026 - Aug 2026

- Architected an event-driven telemetry backend across **EC2, SQS, Lambda, and RDS**, processing **100K+ records/race** with **sub-300ms** analytical queries.
- Partitioned the live data stream across **3 SQS queues** with dedicated Lambda consumers for timing, telemetry, and session events, persisting each workload independently to PostgreSQL.
- Engineered a routed **LangGraph** service over MongoDB Atlas RAG and PostgreSQL text-to-SQL with self-validating rewrite loops, reducing hallucinated responses by **35%**.

Redis-Compatible Storage Engine | Go, TCP, Concurrency

Sept 2026

- Developing an in-memory key-value server in **Go** with RESP over raw TCP, concurrent client handling, command dispatch, and TTL expiration.

EDUCATION

McMaster University

Hamilton, ON

B.Eng. Software Engineering CO-OP

Expected Apr 2028

- Coursework:** Databases, Data Structures & Algorithms, Operating Systems, Software Architecture